The New Focus on Energy Storage Systems Safety

AI OPEN DATA CENTER: Power Distribution

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From fragmented requirements to consensus best practice for data centers

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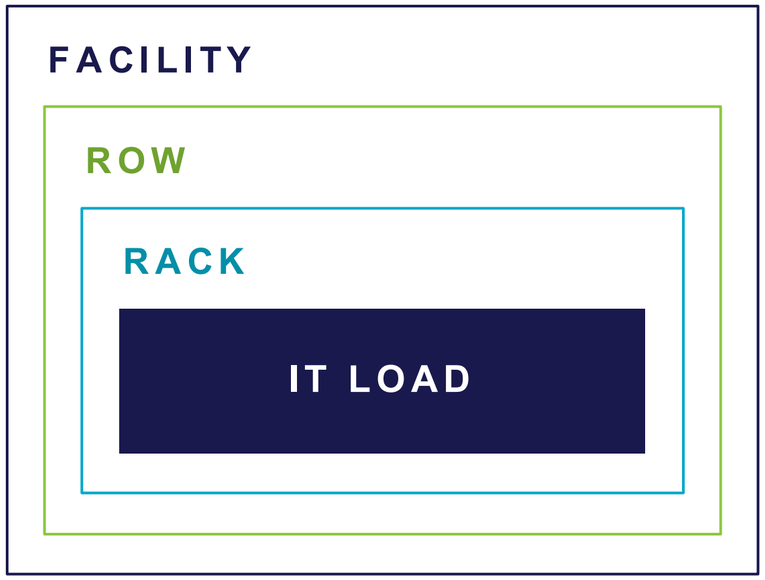
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ESS is no longer confined to the UPS battery room

ESS is now distributed across the data-center power architecture



Facility-level ESS

Centralized UPS and BESS

High stored energy

Building and fire-code interaction

Row-level ESS

Shared backup or sidecar systems

Closer integration with power distribution

Tighter space and ventilation constraints

Rack-level ESS

Battery backup units and distributed storage

Closest to the IT load

High density and service-access challenges

Technologies include: lithium-ion, nickel-zinc, lead-acid, sodium-ion and supercapacitors.

The technology may be similar. The installation context—and therefore the safety case—is not.

Product certification is only one part of deployment approval

Four layers determine whether an ESS deployment is acceptable

01	**Product**	Cell, module and system certification
02	**Application**	Topology, duty cycle, fault energy and control strategy
03	**Installation**	Location, quantity, separation, ventilation and fire protection
04	**Acceptance**	Local code, authority having jurisdiction, owner and insurer requirements

- **A valid certification may still be insufficient if the test scope, installation configuration or jurisdiction differs.**

The rules overlap, but they do not always align

Where requirements generally agree

- Electrical hazards must be controlled
- Failure propagation must be evaluated
- Off-gassing and fire consequences must be addressed
- Detection, protection and emergency response are required

Where requirements differ

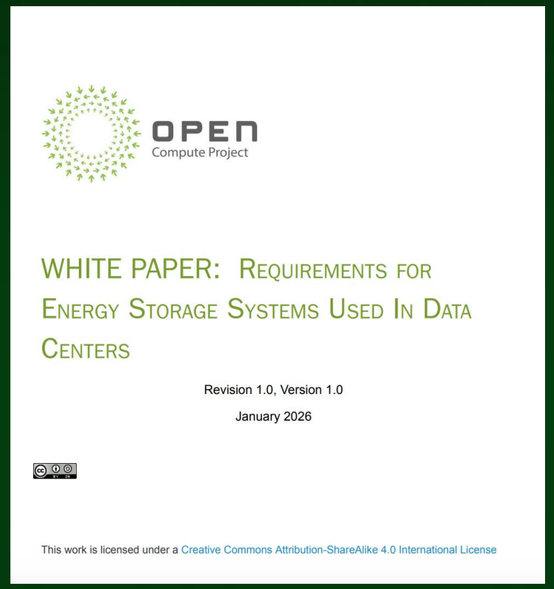
- ESS definitions and classifications
- Test methods and acceptance thresholds
- Separation distances and quantity limits
- Ventilation, suppression and enclosure requirements
- Indoor versus outdoor treatment

Where equivalence breaks down

- Evidence accepted in one region may not be accepted in another
- Similar certifications may not be treated as equivalent
- Insurer requirements may go beyond statutory code
- A certified system may require reassessment after integration

- **These conflicts are often discovered late—during permitting, insurer review or commissioning.**

The White Paper creates a common starting point



RECENTLY PUBLISHED

Requirements for Energy Storage Systems Used in Data Centers

The White Paper helps the industry understand:

- Where the ESS is deployed: facility, row or rack
- Which energy-storage technology is being considered
- Which electrical, thermal, gas, fire and chemical hazards apply
- Which codes, standards, certifications and insurer requirements are relevant
- What information is needed to make an informed deployment decision

Download the White Paper

- **The White Paper organizes the requirements landscape; it does not replace local codes or the authority having jurisdiction.**

Workstreams

ESS PROJECT ORGANIZATION

Organized intentionally so the result is usable and actionable.

1 ESS Technical Reference (WS- V2)

PURPOSE

Develop the content and hold it in one reference document

COVERS

Applications
Technologies
Standards
Terminology

OUTPUT

A whitepaper that serves as the technical reference

2 ESS Selection & Deployment Framework

PURPOSE

Enable implementation: structure the content and guide its use

COVERS

Framework structure
Selection and adoption
Deployment requirements

OUTPUT

A roadmap that points into the whitepaper for the detail

Workstream 1 : ESS Technical Reference (WS- V2)

Objective: a whitepaper that serves as the technical reference for ESS in data centers

Workstream 1 develops the bulk of the content and the whitepaper that contains it. It is a V2 of the existing whitepaper

01	**Applications**	How energy storage will be used in data centers: the roles it can play, the use cases, and an initial characterization of the load profile.
02	**Technologies**	The characteristics of the different ESS technologies — batteries, capacitors and others
03	**Standards**	Built on V1's in-depth coverage, expanding the international picture and standards still to come, and aimed at helping the reader navigate the process. Scope to be refined as a group.
04	**Terminology**	A defined set of terms carried through the whitepaper, to give a consistent reference point and support industry alignment.

- **Written so applications and technology capabilities can be mapped to each other, not just described.**

Workstream 2 : ESS Selection & Deployment Framework

The result: a guide to ESS technology selection and deployment in data center facilities

Workstream 2 establishes the framework that connects the applications, technologies and standards content of the whitepaper

01	**Framework**	The connecting structure between applications, technologies and standards — how that content is defined and built, and how the pieces map to one another. This is the primary work product of the workstream.
02	**Selection and adoption**	A roadmap from a defined need to a selected technology, using existing frameworks such as IEEE 1679 or IEC 62933 whenever they apply, and referencing the whitepaper for the underlying detail.
03	**Deployment requirements**	What it takes to actually deploy and run these systems in a data center: integration, commissioning, installation, maintenance and monitoring, together with capability requirements such as ramp rate, so the technology works with data center loads.

- **Workstream 2 is the connecting point — it points into the whitepaper for the detail rather than repeating it.**

We need the cases where the rulebook stops being clear

Contribute your experience

We are looking for:

- Real deployment architectures and use cases
- Conflicting regulatory or insurer requirements
- Certifications or test evidence that were not accepted
- Installation and commissioning lessons
- Incident, failure and mitigation experience
- Regional requirements that others may not know

Who should participate?

Operators and hyperscalers • ESS manufacturers • Standards organizations and laboratories • AHJs and fire-safety professionals • Insurers • Designers and integrators

If energy storage touches your data center, your experience belongs in this work.

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CALL TO ACTION

Join the OCP Energy Storage workstreams

Review the published White Paper

- **Contribute a deployment case or regulatory challenge**